TEXORA · LECTURE SHEET

Nanocomposite Fibres and Coatings; Nanofibres

Two ways nanoscale structure becomes a textile product: nanofillers dispersed inside fibres and coatings — delivering gas barrier, antimicrobial and conducting functions — and fibres themselves spun at nanoscale, opening filtration and tissue engineering.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 8 · LEVEL 4 TERM 2

1. Nanocomposites — the idea in one line

A **nanocomposite** disperses a nanoscale filler in a bulk matrix so that properties change at loadings of only a few percent. The mechanism is interface: a few percent of nanoclay platelets presents hundreds of square metres of filler surface per gram of composite, and everything the filler does — blocking gas, stiffening, charring, conducting — happens at that interface. The enemy, from Topic 7, is agglomeration: undispersed filler is just expensive sand. In textile form, nanocomposites appear as **fibres** (filler inside the spun polymer) and as **coatings** (filler in a surface layer).

2. Nanocomposite fibres

- **Route:** nanoparticles or nanoclay are compounded into the polymer (melt mixing, or masterbatch dilution) and the filled polymer is melt- or solution-spun like a normal fibre.
- **What it buys:** functions that survive the fibre's whole life because they are *through* the fibre, not on it — barrier fibres, flame-retarded fibres, conductive fibres (carbon black, CNT, graphene loadings near the percolation threshold), antimicrobial fibres (silver, ZnO inside).
- **The costs:** filler raises melt viscosity, can block spinnerets and filters, may reduce drawability and tenacity if poorly dispersed, and buries much of the active material where a surface function cannot use it (the YM-4203 sheet's in-fibre trade-off). Loading is therefore a compromise: enough for function, low enough to spin.

3. Nanocomposite coatings — and the barrier worked example

In a coating, platelet fillers such as montmorillonite nanoclay align roughly parallel to the film. A gas molecule can no longer travel straight through: it must wander around platelet after platelet — the **tortuous path**. The Nielsen model estimates the effect:

$$P/P_0 = (1 - \varphi) / (1 + (L/2W) \cdot \varphi)$$

where φ is the filler volume fraction and L/W the platelet aspect ratio.

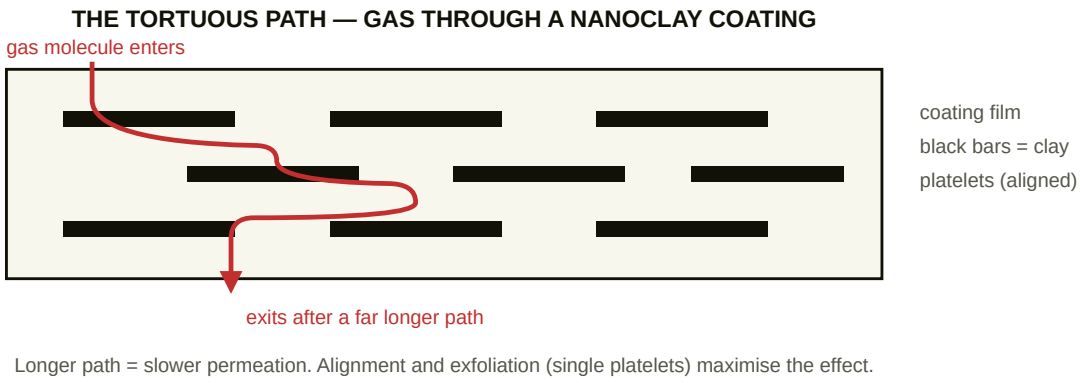


Figure 1 — Tortuous-path barrier: aligned nanoclay platelets force diffusing gas onto a winding route through the coating.

WORKED EXAMPLE — NIELSEN BARRIER ESTIMATE

Nanoclay aspect ratio L/W = 100, loading φ = 0.03 (3 vol%):

$P/P_0 = (1 - 0.03)/(1 + (100/2) \times 0.03) = 0.97/(1 + 1.5) = 0.97/2.5 \approx$ **0.39**.

Three percent of well-exfoliated clay cuts gas permeability to about 39% of the unfilled coating — the nanocomposite bargain in one line: property change out of all proportion to loading, *provided* the platelets are separated and aligned; the same clay agglomerated into stacks behaves as if its aspect ratio were a tenth as large and delivers a fraction of this.

Other coating functions (the syllabus's examples): **antimicrobial** coatings (silver, ZnO, MgO particles in a binder — surface-placed, so loadings are low and action immediate, with durability the governing question); **conducting** coatings (CNT/graphene/silver networks for antistatic, heating and sensing layers); **flame-retardant** coatings (clay and oxide fillers promoting char and blocking heat feedback).

4. Nanofibres — preparation

A **nanofibre** is a fibre below about 1,000 nm diameter, typically 100–500 nm. Preparation routes:

ROUTE	METHOD IN BRIEF	CHARACTER
Electrospinning	High voltage draws a fine jet from a polymer solution/melt; the jet whips, thins and dries into nanofibres collected as a web (detailed in the YM-4203 electrospinning sheet)	The laboratory and industrial standard; huge polymer choice; low throughput per needle, scaled by multi-needle and needleless systems

Sea–island (bicomponent) splitting	A fibre of many fine “island” filaments in a soluble “sea” is spun conventionally, then the sea is dissolved	Uses spinning-mill logic; fibres typically reach the fine microfibre range
Melt blowing (fine settings)	Hot air attenuates molten polymer into micro/nano fibres	Industrial webs for filtration; broader diameter distribution
Forcespinning / centrifugal	Centrifugal force throws solution jets from a rotating head	No high voltage; higher throughput; coarser control
Phase separation / self-assembly / drawing	Gel freezing and extraction; molecular assembly; direct drawing from solution	Specialist and research routes (notably for scaffolds)

5. Nanofibre properties — the surface calculation

For cylindrical fibres the specific surface area is SSA = 4/(ρd) (Topic 2). Applying it across the size range shows what “nano” buys:

WORKED EXAMPLE — SURFACE AREA ACROSS THE FIBRE SCALE

Electrospun PAN nanofibre, d = 200 nm, ρ = 1,180 kg/m³:

SSA = 4/(1,180 × 200 × 10^{-9}) = 4/(2.36 × 10^{-4}) ≈ 16,949 m²/kg ≈ **16.9 m²/g**.

Ordinary PET fibre, d = 15 µm, ρ = 1,380 kg/m³:

SSA = 4/(1,380 × 15 × 10^{-6}) ≈ 193 m²/kg ≈ **0.19 m²/g**.

Ratio ≈ 16,949/193 ≈ **88** — the nanofibre web offers nearly ninety times more surface per gram: the physical basis of its filtration efficiency, adsorption capacity and cell-friendliness.

Alongside surface area come **small pores** (the web's pores are sub-micrometre — fine enough to stop bacteria and fine aerosols while passing air), **low areal weight** (a functional layer can weigh a fraction of a gram per square metre, laminated onto a support), and **fragility** — nanofibre webs are delicate and normally used supported or sandwiched, a design fact, not a defect.

6. Nanofibre applications

- **Filtration** — the flagship use: air filters (including respirator media) and liquid filters capture fine particles at low pressure drop, because the fine fibres intercept particles efficiently while the web stays open. Efficiency rises without the energy penalty of a denser conventional medium.
- **Tissue engineering and wound care** — electrospun scaffolds mimic the scale of the extracellular matrix (whose own fibres are nanoscale), so cells attach and grow along them; wound dressings use the same webs for exudate handling and barrier action.

- **Protective clothing** — thin nanofibre membranes block aerosols and liquid-borne agents while transmitting moisture vapour: protection that breathes.
- **Others** — battery separators, sensors, drug-delivery patches, and reinforcement veils in composites.

Glossary

Exfoliation — separation of clay stacks into individual platelets in a polymer. **Percolation threshold** — the loading at which conductive particles first form a connected network. **Tortuosity** — the ratio of the actual path length through a medium to the straight-line distance. **Precursor** — the material a process starts from (e.g. the polymer solution for electrospinning). **Scaffold** — a temporary structure that supports cell growth in tissue engineering.

Key points

• Nanocomposites work by interface: a few percent of well-dispersed filler changes barrier, stiffness and flame behaviour; agglomeration destroys the bargain.
• In-fibre nanocomposites are durable but bury actives; coatings place actives at the surface, with durability to prove.
• Nielsen's model: 3 vol% aligned clay of aspect ratio 100 cuts permeability to ≈ 0.39 of the unfilled film.
• Nanofibres come mainly from electrospinning (with bicomponent, melt-blown and centrifugal alternatives) and deliver ≈ 88× the surface per gram of a 15 µm fibre at 200 nm.
• Filtration, tissue engineering and breathable protection are the governing nanofibre applications; webs are used supported because they are delicate.

Practice

1. Repeat the Nielsen estimate for $\varphi = 0.05$ and aspect ratio 200, and for the same φ with poorly exfoliated clay of effective aspect ratio 20. What does the comparison teach?
2. Calculate the SSA of a PLA nanofibre of diameter 300 nm ($\rho = 1{,}240$ kg/m³) intended for a tissue scaffold, and compare it with the PAN example.
3. A respirator medium must gain particle efficiency without raising breathing resistance much. Explain in three sentences why a thin nanofibre layer on a conventional support is the standard answer.

Answer key

1. $\varphi = 0.05$, L/W = 200: $P/P_0 = 0.95/(1 + 100 \times 0.05) = 0.95/6 \approx$ **0.16**. Aspect ratio 20: $0.95/(1 + 10 \times 0.05) = 0.95/1.5 \approx$ **0.63**. Lesson: dispersion quality (effective aspect ratio) is worth more than

loading — four times the barrier difference at identical composition.

2. SSA = $4/(1{,}240 \times 300 \times 10^{-9}) = 4/(3.72 \times 10^{-4}) \approx 10{,}753$ m²/kg ≈ **10.8 m²/g** — about two-thirds of the 200 nm PAN value, as expected since SSA scales as 1/d (and slightly with ρ).
3. The nanofibre layer's sub-micrometre fibres intercept fine particles efficiently at the surface, so a very thin, light layer adds large efficiency. Because the layer is so thin, its own air resistance stays small; the conventional support provides mechanical strength and the bulk of the flow path at low resistance. Efficiency is bought with surface physics, not with thickness — so breathing resistance rises far less than it would by densifying a conventional medium.
